

Question: 1

Install a CPU and a RAM stick onto a motherboard, then apply thermal paste using the pea-sized dot technique before mounting the CPU cooler.

Hint:
Take a clear photo of each step as you complete it for your documentation.

Answer: 1

Step 1: Installing the CPU



1. Carefully aligning and inserting CPU

Step 2: Installing the RAM Stick



2. Pressing RAM into slot until it clicks

Step 3: Applying Thermal Paste



3. Pea-sized dot of thermal paste

Step 4: Mounting the CPU Cooler



4. Lowering the cooler bracket onto CPU

Question: 2

Mount the motherboard inside a PC case and connect both the 24-pin ATX and 8-pin CPU power connectors from the power supply, making sure each connector clicks into place.

Answer: 2

Steps to Mount the Motherboard and Connect Power

Align with I/O Shield: Slide the motherboard gently into the PC case, making sure the rear ports (USB, audio jacks) line up perfectly with the I/O shield opening at the back.

Secure with Screws: Align the motherboard screw holes with the brass standoffs inside the case. Tighten the screws using a Phillips-head screwdriver to secure the board firmly.

Plug the 24-Pin ATX Cable: Locate the largest wide power cable from the power supply. Align it with the 24-pin slot on the right side of the motherboard and press down firmly until the plastic safety latch clicks into place.

Plug the 8-Pin CPU Cable: Find the 8-pin (or 4+4 pin) EPS power cable. Connect it to the dedicated CPU power slot, usually located at the top-left corner near the processor socket, until it firmly clicks locked.

Motherboard Installation and Power Connections

Step 1: Mounting the Motherboard



1. Screwing the motherboard to the standoffs.

Step 2: Connecting 24-pin ATX Power



Step 2: Connecting 24-pin ATX Power



2. Inserting 24-pin ATX power cable.

Step 3: Connecting 8-pin CPU Power



3. Securing the 8-pin CPU power connector (Latch 'clicked').

Question: 3

Install a storage drive (SSD or HDD) and a GPU (graphics card) into the appropriate slots, then connect all necessary data and power cables to each device.

Answer: 3

Steps to Install Storage Drive and GPU

Install the Storage Drive (SSD/HDD): Slide your 2.5-inch SSD or 3.5-inch HDD into an empty drive bay cage inside the computer case. Secure it using mounting screws or snap it into the tool-less plastic bracket.

Connect Storage Cables: Plug a flat **SATA Data Cable** from the drive into a SATA port on the motherboard. Then, plug a wider **SATA Power Cable** from the power supply unit (PSU) into the back of the storage drive.

Install the GPU (Graphics Card): Open the locking latch on the long **PCIe x16 slot** on your motherboard. Align the graphics card with the slot and push it straight down until you hear a clear click sound indicating the slot latch has locked it in place. Secure the card's metal bracket to the PC case frame with screws.

Connect GPU Power Cables: Locate the **6+2 pin PCIe power cable(s)** coming from your power supply unit. Push them firmly into the matching power ports on the top edge side of the graphics card until they click securely.

Question: 4

Organize and secure all cables inside the case using zip ties or Velcro straps to ensure good airflow and a tidy appearance.

Hint: Take a photo showing your final cable management from both the front and back of the case.

Answer: 4

Steps for Cable Management

- **Route Cables Through Cutouts:** Pull all major power and data cables from the front chamber through the rubber grommets or routing holes directly to the back of the case.
- **Bundle Similar Cables:** Group thick cables (like the 24-pin ATX) together, and tie down thinner wires (like fan headers or front panel cables) along the built-in channels.
- **Secure with Straps:** Use **Velcro straps** or **zip ties** to tie the cable bundles tightly to the case's built-in anchor loops, keeping them completely flat against the frame.
- **Clear the Airflow Path:** Double-check that no stray wires are hanging in the front compartment near the fans or components to ensure unobstructed cooling airflow.



Question: 5

Perform a final verification: check for any loose screws, confirm all power/data connections are firmly attached, and document your completed build with a series of labeled photos showing each major component installed.

Answer: 5

Final Verification Checklist

- **Hardware Fasteners:** Checked all motherboard standoffs, GPU brackets, and drive mountings to ensure no loose screws are left vibrating inside the chassis.
- **Power Connection Security:** Verified that the 24-pin ATX, 8-pin CPU, and PCIe power connectors are securely plugged into their sockets with their plastic retention clips completely locked.
- **Data Cables:** Confirmed that all SATA data cables are fully pushed into the storage drives and motherboard ports.
- **Fan & Front Panel Headers:** Double-checked that the CPU cooler fan, case fans, power button, and front USB headers are connected to the correct pins.

